MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548086

(INTEGRATOR, RATE)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the rate integrator assembly used in the Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50652	Circuit Card Assemblies, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

F10548086	Integrator, Rate

PACKAGING DATA SHEET

10548086	Packaging of Rate Integrator

FSC 1285

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(Copies of specifications, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10548086 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the require - ments of MIL-C-50652.

3.3 Environmental service conditions. - As specified in MIL-C-50652.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the rate integrator.

3.5 Performance. - For the requirements of 3.5.1 through 3.5.7 logic and input and output levels shall be as follows:

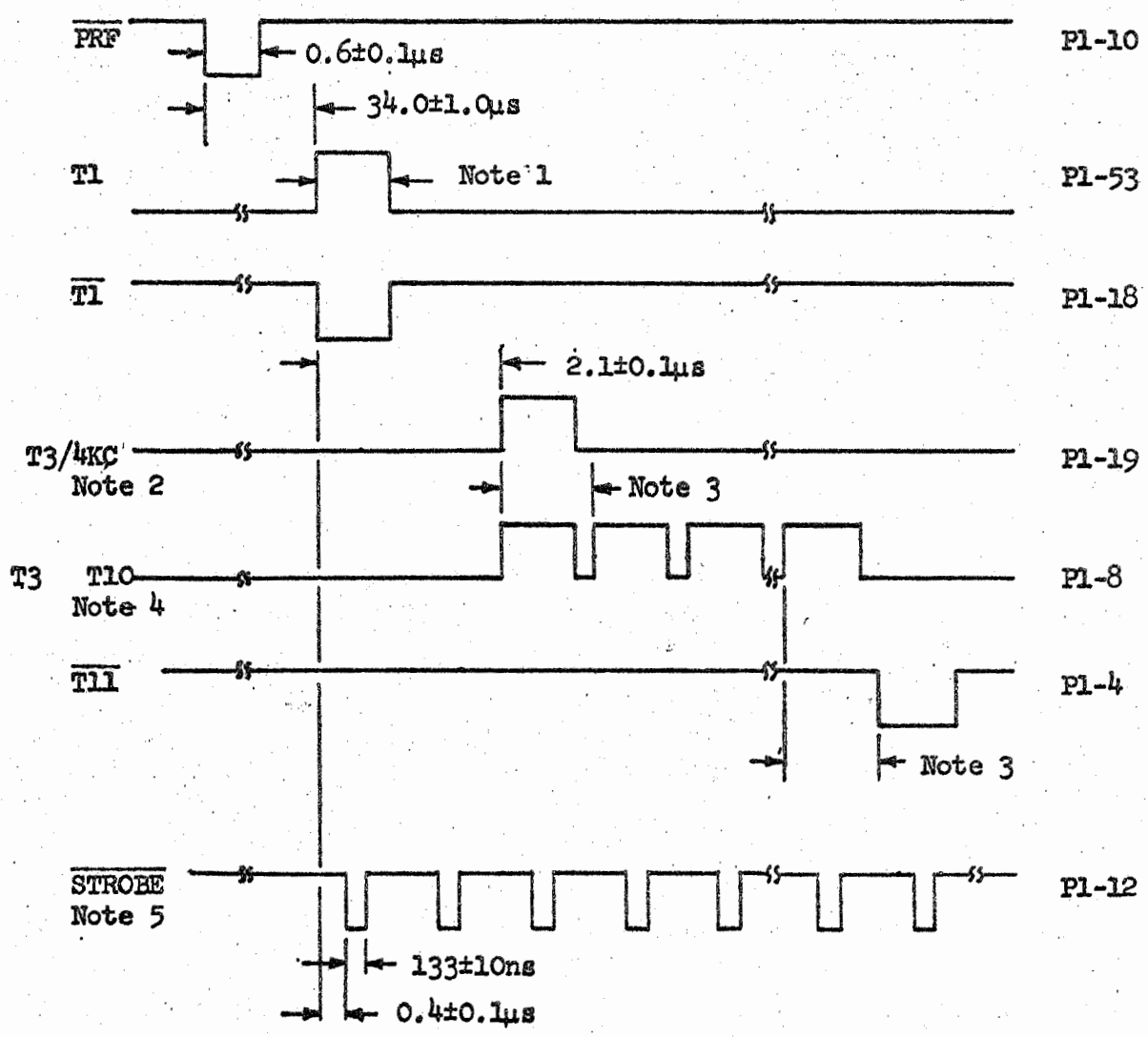
Logic 0 — 0.0 ±0.5 Vdc level
Logic 1 — +3.5 ±1.0 Vdc level

3.5.1 Input power. - The input power applied between terminals P1-30, P1-31, and P1-32 (+5 V) and terminals P1-24, P1-25, and P1-26 (+5 V return) shall be 5.0 ±0.3 volts direct current (Vdc), at 200 milliamperes (mA), regulated; with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.2 Input signals. - Input signals to the rate integrator assembly shall be $\overline{\text{PRF}}$, T1, $\overline{\text{T1}}$, T3/4KC, T3→T10, $\overline{\text{T11}}$, and $\overline{\text{STROBE}}$ as shown in Figure 1. A positive pulse shall be a change in direct current (dc) level from 0.0 ±0.5 volts (V) to +3.5 ±1.0 V and return to 0 V. A negative pulse shall be a change in dc level from +3.5 ±1.0 V to 0.0 ±0.5 V and return to +3.5 V.

3.5.3 $\dot{\overline{R}}_S$ output. - With a logic 1 level applied to P1-51 ($\dot{R}$ RESET input), the output signals at P1-50 and P1-22 ($+\dot{\overline{R}}_S$ and $-\dot{\overline{R}}_S$ output signals respectively) shall be complementary and change logic levels each time a logic level transition from 1 to 0 is applied to P1-49 ($\dot{R}$ OVF input). With the output signals at P1-50 and P1-22 at logic levels of 0 and 1 respectively, and the logic signal applied to P1-51 is changed from a 1 to a 0 level, the output signals shall change to a logic 1 at P1-50 and logic 0 at P1-22. With a logic 0

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NOTES:
1. All T pulses: duration 0.8 ±0.1 µs @ 20 kHz PRF except T3/4KC.
2. T3/4KC pulse 0.8 ±0.1 µs at 4 kHz PRF.
3. All T pulses spaced 1.06 ±0.1 µs leading edge to leading edge.
4. T pulses 3 through 10 in serial form.
5. Train of 15 pulses, spaced same as T pulses and coincident with T pulses.

FIGURE 1. INPUT SIGNALS.

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level applied to P1-51 the output signals at P1-50 and P1-22 shall not change levels when logic level transitions are applied to P1-49.

3.5.4 Strobe output. - The STROBE output signal at P1-41 shall be a train of 15 complementary pulses to the $\overline{\text{STROBE}}$ input signal shown in Figure 1.

3.5.5 $-\Delta$ R SIGN output. - With the input and output logic level conditions sequentially established as specified in Table I. the $-\Delta$ R sign output signal at P1-3 shall be as specified in Table I. The logic levels at P1-22 are derived as specified in 3.5.3. The T11 output pulse shall be complementary to the $\overline{\text{T11}}$ pulse shown in Figure 1.

TABLE I. - $-\Delta$ R SIGN OUTPUT

SIGNAL CONDITIONS			OUTPUT SIGNAL
STEP	P1-51	P1-22	P1-3
1	Logic 0	Logic 0	Logic 0
2	Logic 1	Logic 0	T1 pulse as shown in Figure 1.
3	Logic 1	Logic 1	T1 and T11 pulses as shown in Figure 1.

3.5.6 Track output. - The track signal output at P1-39 shall have the same logic level as is applied to P1-47.

3.5.7 Δ R output. - With the input logic levels applied sequentially as specified in Table II, and the timing signals applied to P1-20 as specified in Table II, the Δ R output signal shall be as specified in Table II. Timing signals T3 $\longrightarrow$ T5 shall be T pulses T3 through T5 in serial form with the same characteristics as T pulses shown in Figure 1. Timing signals T6 $\longrightarrow$ T10 shall be T pulses T6 through T10 in serial form with the same characteristics as T pulses shown in Figure 1. The output signal condition at P1-50 is determined as specified in 3.5.3.

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TABLE II. - Δ R OUTPUT

	INPUT SIGNALS			OUTPUT SIGNALS	
STEP	P1-2	P1-47	P1-20	P1-50	P1-43 (See note)
1	Logic 0	Logic 0	-	Logic 1	T1 strobe pulse at 4 kHz PRF.
2	Logic 0	Logic 1	T6, T10	Logic 1	T11 strobe pulse at 20 kHz PRF.
3	Logic 0	Logic 1	T3, T5	Logic 0	T11 strobe pulse at 20 kHz PRF.
4	Logic 1	Logic 1	-	Logic 0	All 155 strobe pulses at 20 kHz PRF.

NOTE: Individual STROBE pulses shall be complementary to the STROBE pulses shown in Figure 1 and occurring at T periods indicated.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. - As specified in MIL-C-50652.

4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special sampling. - As specified in MIL-C-50652.

4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.2.3.

4.5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652.

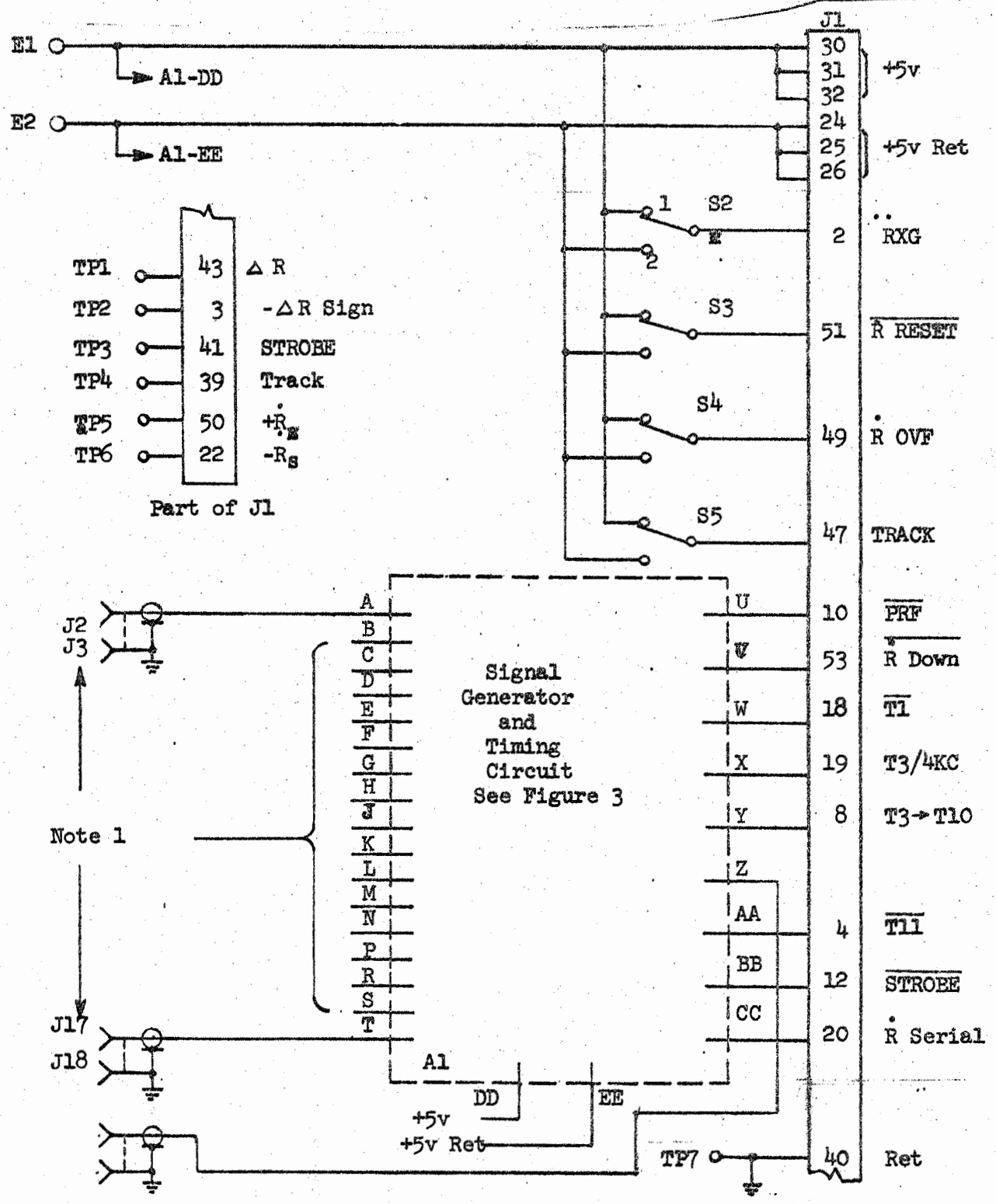
4.5.3 Test equipment. - In conjunction with 4.5 the following test equipment listed in Table III shall be utilized in the performance of the applicable tests as specified in 4.6.3.

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TABLE III. - TEST EQUIPMENT LIST

ITEM NUMBER		DESCRIPTION
1.	Power supply	5.0 ±0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity, 300 mA.
2.	Pulse generator	Primary or channel 1 pulse output, +5 V peak into 50-ohms at 20 kHz PRF, 0.8 ±0.1 μs duration. Delayed sync on channel 2 pulse output, +5 V peak into 50-ohms with variable delay from channel 1 of 34.0 ±1.0 μs.
3.	Data generator (digital word generator)	15 parallel channels, minimum of 15 selectable bits per channel, with 1/0 content of each bit independently programmed. Data output shall be of return to zero (RZ) with controllable pulse width of 0.8 ±0.1 μs. PRF shall be externally triggered from a 20-kHz pulse. Output pulse amplitude, 5 V into a 50-ohm load.
4.	Pulse generator	Positive pulses; 5 V peak into a 50-ohm load, 133 ±10 ns duration, for each external trigger pulse applied.
5.	Oscilloscope	Dual channel; rise time of scope plus measuring probe, 10 ns or less; sweep period 50 ns/cm or greater, input resistance 8 megohms or greater; input capacitance, 25 pF or less.
6.	Test fixture	See Figure 2. See Table IV for test fixture parts list.

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Note 1. J3 through J17 connections are listed in Table VI.

FIGURE 2. TEST FIXTURE SETUP.

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TABLE IV. - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
A1	Timing circuit, see Figure 3, See Table V for parts list.
E1	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector; any of the following or equivalent:
	(1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois.
	(2) 500946-01, National Connector Corp. Minneapolis, Minn.
	(3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
J2 thru J19	Connector, receptacle, UG-25B/U or equivalent.
S1 thru S5	Switch SPDT.
TP1 thru TP6	Tip jack, H.H. Smith 1501 or equivalent.

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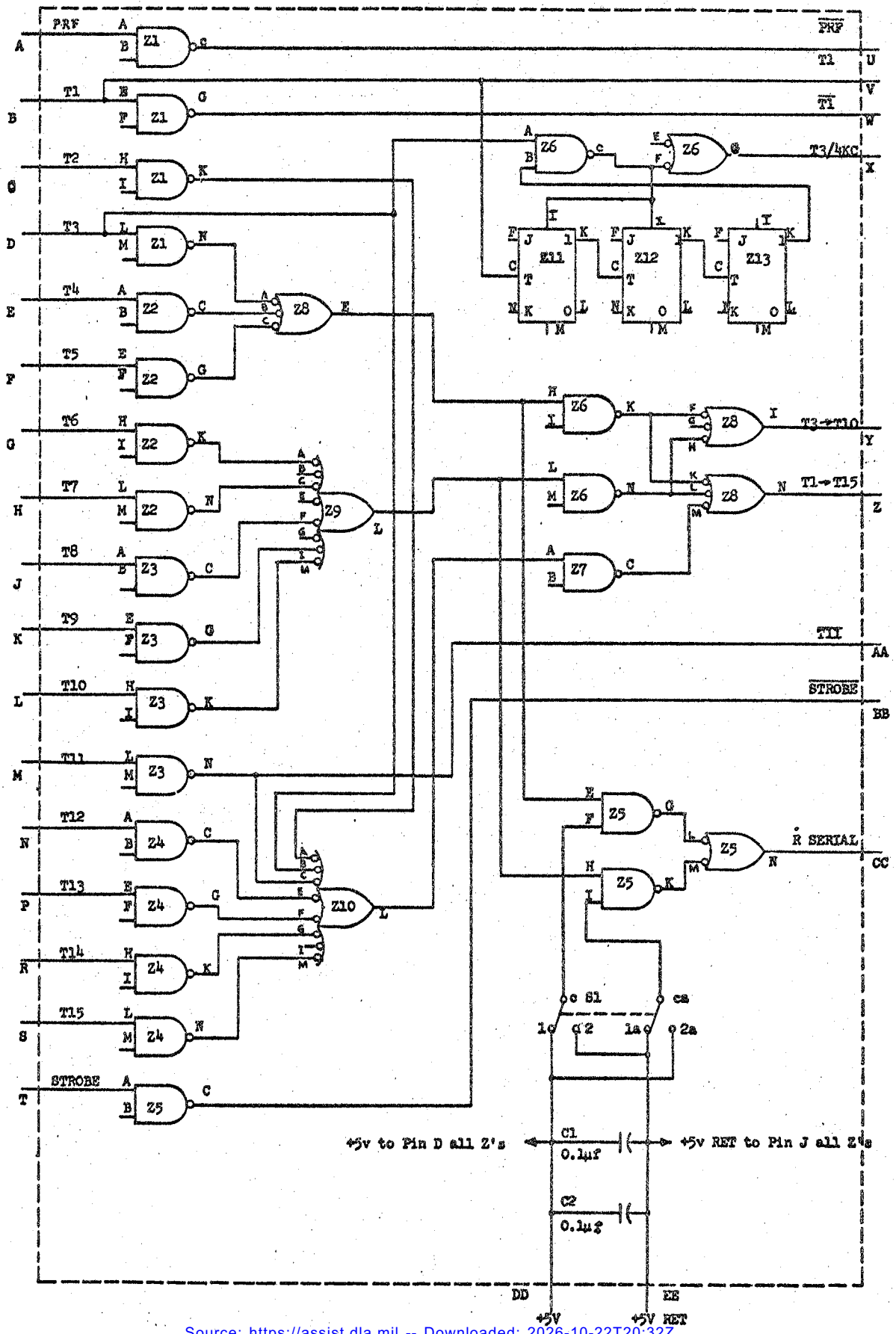


FIGURE 3. SIGNAL GENERATOR AND TIMING CIRCUIT.

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TABLE V.-SIGNAL GENERATOR AND TIMING SIGNAL CIRCUIT PARTS LIST

ITEM	DESCRIPTION
C1	Capacitor 0.1 μF @ 50 WVdc; SSM-1-189 Gulton Industries, Inc. or equivalent.
C2	Same as C1.
Z1 thru Z7	Integrated circuit, SG-140-37 Sylvania or equivalent.
Z8	Integrated circuit, SG-190-21 Sylvania or equivalent.
Z9	Integrated circuit, SG-60-21 Sylvania or equivalent.
Z10	Same as Z9.
Z11 thru Z13	Integrated circuit, SF-50-28 Sylvania or equivalent.

4.6 Test methods and procedures.

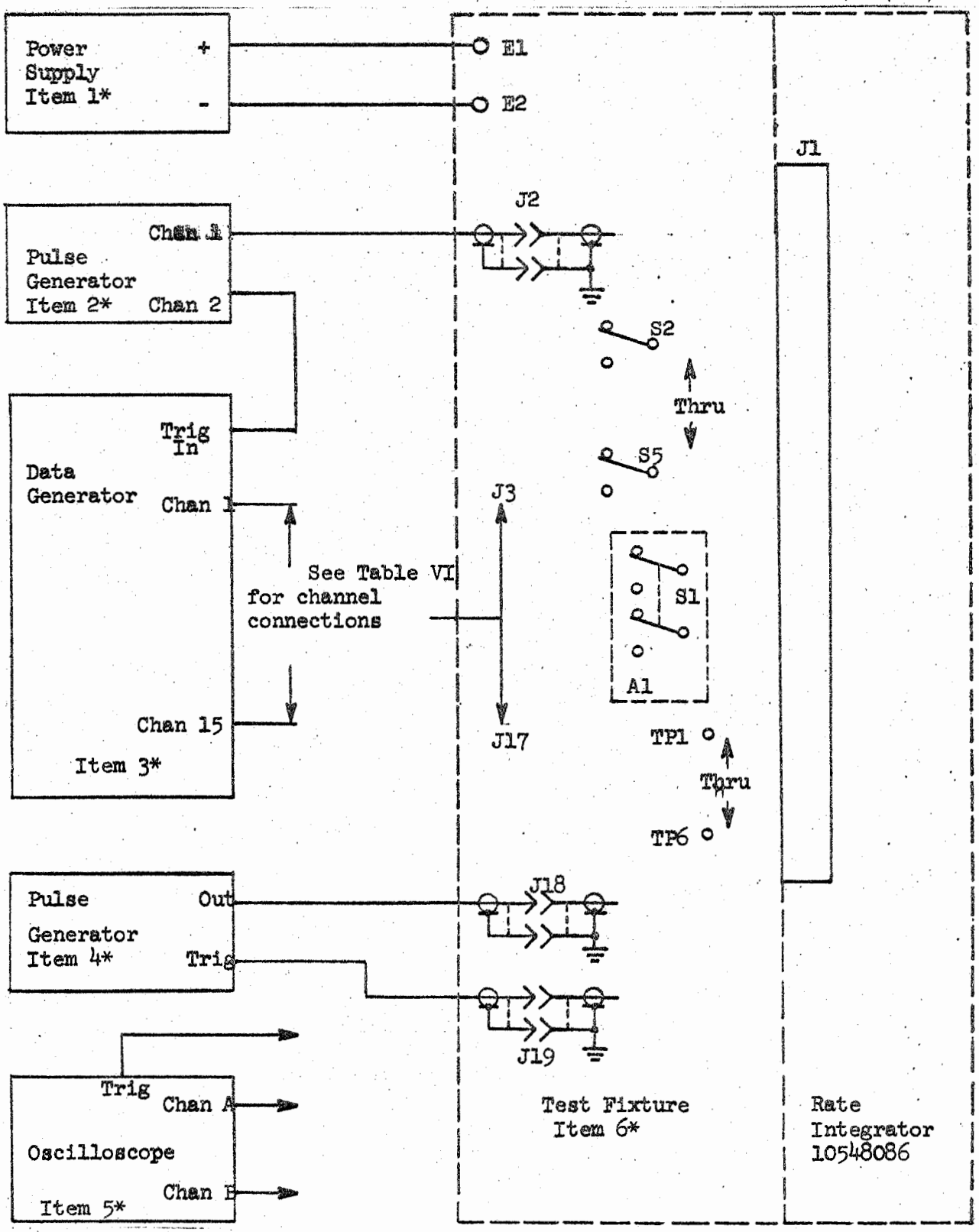
4.6.1 Environmental. - As specified in MIL-C-50652.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. - The selected rate integrator circuit card assemblies shall be tested as follows: Set up equipment as shown in Figure 4. Use oscilloscope, item 5, for observation of input and output signals specified in 4.6.3.2 through 4.6.3.6. When specified, a logic 0 shall be observed as a 0.0 ±0.5 Vdc level and a logic 1 as a +3.5 ±1.0 Vdc level. Switches S2 through S5, on the test fixture, item 6, are used to supply the logic level input signals specified in 4.6.3.3 and through 4.6.3.7. These switches supply a logic 1 in position 1 and a logic 0 in position 2.

4.6.3.1 Input power. - Using item 1, apply input power as specified in 3.5.1.

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*Refer to 4.5.2.3

FIGURE 4. TEST SETUP.

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4.6.3.2 Input signals. - Use pulse generator, item 2, to supply the complement of $\overline{PRF}$ signal, as shown in Figure 1, to J2 on the test fixture, item 6. Use data generator, item 3, to supply the timing pulses T1 through T15 with conditions as specified in notes 1 and 3 of Figure 1. Connections for the output channels of the data generator, item 3, are specified in Table VI. Use T1 through T15 in serial form provided as an output signal at J18 on the test fixture, item 6, as the trigger source to pulse generator, item 5. Use the output of pulse generator, item 5, to apply the complement of the STROBE signal (train of 15 pulses) as shown in Figure 1 to J19 on the test fixture, item 6.

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TABLE VI. - TIMING SIGNAL CONNECTIONS

TIMING SIGNAL	DATA GENERATOR CHANNEL	TEST FIXTURE JACK	A1 CIRCUIT TERMINAL
T1	1	J3	B
T2	2	J4	C
T3	3	J5	D
T4	4	J6	E
T5	5	J7	F
T6	6	J8	G
T7	7	J9	H
T8	8	J10	J
T9	9	J11	K
T10	10	J12	L
T11	11	J13	M
T12	12	J14	N
T13	13	J15	P
T14	14	J16	R
T15	15	J17	S

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4.6.3.3 $\dot{R}_S$ output. - Use switches S3 and S4 on the test fixture, item 6, to apply logic levels as specified in 3.5.3. At TP5 and TP6 observe the $+\dot{R}_S$ $-\dot{R}_S$ output signals as specified in 3.5.3. When using switch S4 to establish logic level transitions at P1-49 ($\dot{R}$ OVF input signal) as specified in 3.5.3. It may be necessary to toggle this switch several times to achieve logic level changes at the output terminals P1-22 and P1-50 due to contact bounce in the switch.

4.6.3.4 STROBE output. - At TP3 observe the STROBE output signal as specified in 3.5.4.

4.6.3.5 $-\Delta$ R sign output. - Using switches S3 and S4 on the test fixture to sequentially establish the signal conditions as specified in Table I, observe the $-\Delta$ R sign output signal at TP2 as specified in Table I.

4.6.3.6 TRACK output. - At TP4 observe the TRACK output signal as specified in 3.5.6.

4.6.3.7 Δ R output. - Use switches S2 and S5 to obtain logic levels at P1-2 and P1-47 respectively as specified in Table II. Use switch S4 as specified in 4.6.3.3 to establish logic levels at P1-50 as specified in Table II. Use switch S1 in position 1 to obtain input signals T3 $\longrightarrow$ T5 and position 2 to obtain input signals T6 $\longrightarrow$ T10 as specified in Table II. Perform the steps sequentially in Table II and observe the Δ R output signal at TP1 as specified in Table II.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548086. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use. - The rate integrator assembly is used in the Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The rate integrator assembly derives incremental range changes (Δ R) from the serial binary representation of range rate ($\dot{R}$ SERIAL) furnished by the range rate counter B assembly. This R signal is in the form of pulses with a repetition rate corresponding directly to the distance traveled by the target in a fixed time interval.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:
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Preparing Activity:
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Project No. 1285-A249

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